March to Glory – Game Design Document

Prototype · Napoleon game idea for Slitherine & DGBA

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> How to read this document: Part 1 (sections 1–10) gives a short overview of every system. Each topic ends with a box like this one, whose links open Part 2 (Appendices A–E) with the full rules, actions and numbers. Every appendix has a link to go back to its overview. All numbers are prototype values.

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1 Overview

A strategic game set in the Napoleonic era, in which the player must balance the army's Power and Morale by making tough choices and tactical moves to achieve victory and become the Emperor of Europe.

Key facts

- **Genre:** turn-based strategy.
- **Setting:** Napoleon's Russian Campaign.
- **The player** does NOT command Napoleon's army directly: the army marches on its own, and the player commands a small group of agents (armed bands, spies and scouts), each with moves to spend every turn.
- **Goal:** keep the army strong enough in Power and Morale to win the battles along its path and conquer the region's big settlement.

Core loop

Each turn, Napoleon's army advances toward the final battle of the level, which happens after a set number of turns. During the advance, minor battles, rest periods and unpredictable situations can occur. As the army advances, the map opens up, revealing more of the surroundings and giving the player more freedom of movement.

- **Pre-battle actions:** manage the troops' Morale and Power. Boost the army, or move behind enemy lines to discover the enemy's vantage points, sabotage its supply lines and cut its chain of command, so the enemy has less chance of victory.
- **During battle:** the more similar the two armies' stats are, the more turns the battle lasts. Each turn the player can use the information gathered before the battle to send agents to demoralize the enemy army or reduce its Power, so the battle is won in fewer turns.
- **After battle:** the army has lost Power, depending on how long the battle lasted. The player chooses whether to let it rest, which recovers a large part of the Power lost over several turns, or to advance at once.

See the pitch presentation for a visual walkthrough of the core loop.

2 Nomenclature

Term	Meaning
Army	Napoleon's army. It moves on its own; the player only decides what it does at crossroads and after battles (see 4 Army controls).
Agent	A unit the player moves directly: armed band, spy or scout (see 5 Agents).
Power	The army's strength: troops and equipment. A whole number.
Morale	The army's mood, fatigue and excitement, from 0% to 100%.
Cell	One hexagon of the map. Movement is counted in cells per turn.
Point of interest	A place on the map where agents or the army can perform an action (see 6 Points of interest).
Settlement	A village or city that can be conquered: small, medium or big. Settlements have a level.
Crossroads	A cell where the player chooses the army's next objective.
Fog of war	The parts of the map the player cannot see. Scouts clear it.
Health	During a battle, an army's remaining strength, as a percentage of its Power.

3 Power and Morale

Every army, the player's and the enemy's, has two stats. Together they decide how much damage the army deals in battle (see 9 Battle system).

Stat	What it is	Goes up with	Goes down with
Power	Troops and equipment; a whole number	Points of interest, agents joining the army, rest after a battle	Battles; enemy agents
Morale	Mood and fatigue; 0%–100%	Points of interest, agents joining the army, rest after a battle	Marching (−5% each turn); enemy agents

- Morale above 50% gives the army a bonus in battle; below 50%, a penalty.
- Marching costs 5% Morale each turn. Morale does not drop during battles or rest.
- Enemy armies have the same stats and can be weakened by agents and points of interest.

Starting values	Power	Morale
Napoleon's army	300	70%
Big settlement	700	85%

In detail: A Power and Morale · Numbers: E Prototype values

4 Army controls

The player does not control the army directly. Each turn it advances one cell along its path, until it reaches a point of interest. At battles and crossroads the player gives it orders through the UI. The army's path must always be visible on the map, with the number of turns left before the next battle or action.

Order	When	Effect
Rest	After a battle	The army stays in place for a fixed number of turns (TBD) and recovers Power and Morale.
Advance	After a battle	The army starts marching at once: no rest bonus, but it can reach the next objective before the enemy reinforces it.
Choose a destination	At a crossroads	The player picks one of two objectives, each with its own bonus. The army cannot go back, so the other objective is lost.

In detail: A.3 Army orders

5 Agents

Agents are the units the player moves directly. To move one, the player selects it and chooses a cell within its range. On a point of interest, an agent can perform an action with a UI button: the action keeps it in place until the next turn, and its bonus is applied at the end of the turn.

Agent	Moves per turn	Role	Joining the army
Armed band	2 cells	Fights behind enemy lines and conquers settlements	+Power
Spy	3 cells	Infiltrates, gathers information and sabotages	+Morale
Scout	4 cells	Clears the fog of war and spots enemies	More vision; in battle, spots a weak point each turn

- Every agent has a level. Actions give experience; higher levels make actions faster or stronger, up to a cap.
- Spies and scouts have no Power or Morale.

> In detail: B Agents · Actions on each point of interest: C Points of interest

6 Points of interest

Points of interest are scattered across the map. An agent or the army can interact with one when it is in the same cell; the action depends on who is there.

Point of interest	Armed band	Spy	Scout	Army
Weapon deposit	Raid: +100 Power	Sabotage: the big settlement loses 100 Power	Gather: +120 Power (2 turns)	–
Rations deposit	Raid: +10% Morale	Poison: the big settlement loses 10% Morale	Gather: +12% Morale (2 turns)	–
Bridge	Ambush	Trap: −30 Power, −10% Morale, enemy stops 1 turn	Rig and destroy	–
Small settlement	Conquer (1 turn)	Convert (3 turns)	Guerrilla: lowers its level	–
Medium settlement	Kamikaze attack	Gather information	–	Battle
Big settlement	– (destroyed)	Gather information	– (destroyed)	Battle (final)
Artillery position	Destroy (1 turn)	Sabotage (3 turns)	–	–
Command camp	Kill the general	Spread disorder (3 turns)	–	–
Enemy armed band	Fight	Infiltrate	Quick attack	–
Enemy army	Suicide attack	Infiltrate	Hit-and-run	Battle

- **Conquered settlements** give the army a bonus every turn. Small settlements can create resources, recruit agents or be reinforced; medium settlements can create resources, train agents, be raided or be reinforced.
- Conquering the big settlement ends the match.

> In detail: C Points of interest (every action and bonus) · Numbers: E Prototype values

7 Enemies

The enemy AI pressures the player: it steals resources, kills agents, sets traps and disrupts the player’s plan.

Enemy	Role	Visible
Russian armed band	Kills agents; defends and reconquers points of interest	Yes
Russian spy	Gathers information; lowers the player’s Power and Morale income	Only if discovered or near a scout
Russian scout	Reports the player’s moves; slows agents and the army	Only if discovered or near a scout
Russian sniper	Defends cells within a cone of vision	Only after it fires, or if discovered by a spy

Enemy	Role	Visible
Russian Dog Master	Hunts agents with two groups of dogs	Yes
Russian trapper	Sets traps and ambushes agents	Only if discovered or near a scout
Russian army	Defends settlements and attacks Napoleon's army; moves 1 cell per turn	Always

7.1 Prototype: Russian armed band

In the prototype the Russian armed band has one simple behavior: when an agent conquers a specific settlement, usually a medium or big one, an armed band spawns and tries to retake it.

- The spawn is triggered once, when the settlement is conquered.
- If the band reaches the settlement, it reconquers it in 1 turn, whatever the settlement's level, and becomes part of it.

In detail: D Enemies · Prototype steps: D.8 Prototype armed band behavior

8 Information system

Spies can uncover dispatches and intelligence that reveal the movements and objectives of enemy forces. For example:

- "Reinforcements will arrive from the west in 5 turns."
- "There are weapon deposits north-west of town X."
- "There is a secret point of interest at these coordinates."

9 Battle system

A battle starts when the player's army arrives in the same cell as an enemy army. It can last several turns: its outcome and length depend on the two armies' Power and Morale.

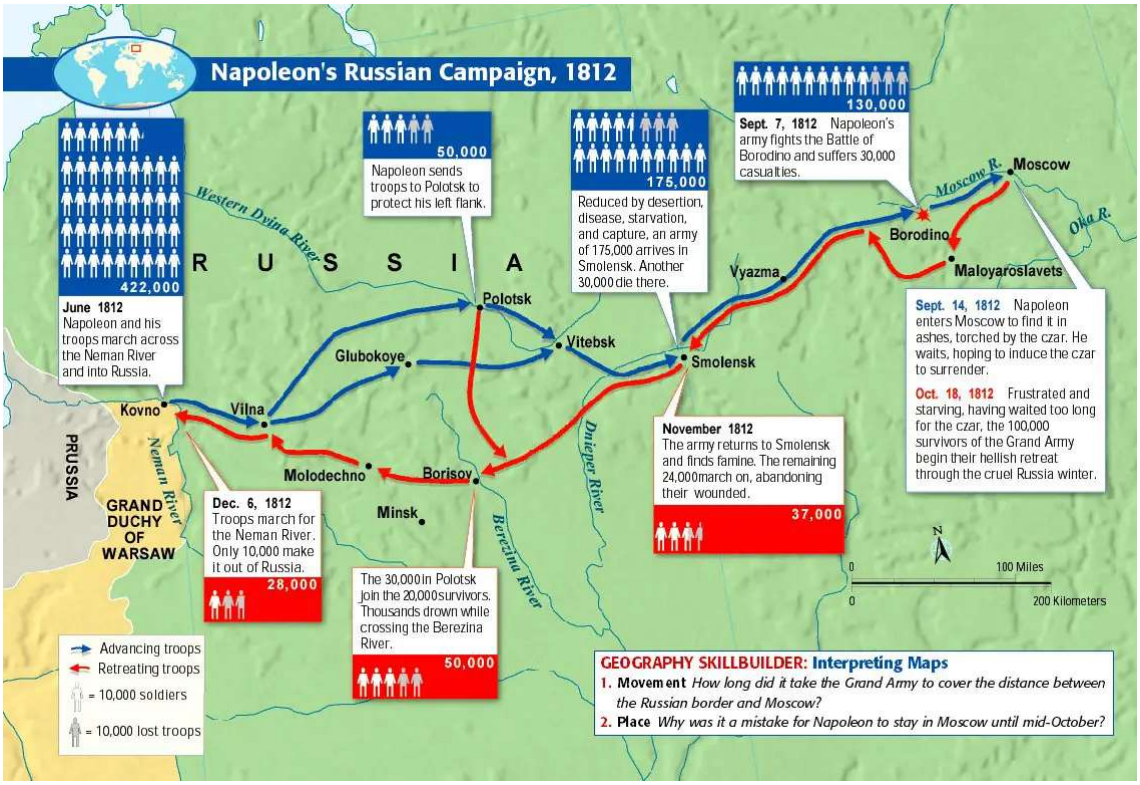
- Each turn, each army deals **Damage per turn** to the other, based on its Power and Morale.
- A message log tells the player how many turns are left before the battle is won or lost; it is updated every turn, so the player sees the effect of their agents' actions.
- The battle is won when the enemy army has no Health left, or when it is routed: a random event, only for Russian armies, that becomes more likely as their Health and Morale drop.

Outcome	Health left	Power lost by Napoleon's army
Great victory	Over 75%	20%
Victory	50%–75%	35%
Suffered victory	25%–50%	50%
Defeat	–	Game over

In detail: E.1 Battle formulas

10 Narrative

Set during the Russian Campaign, the game follows the sequence of battles fought during the campaign. The player must keep up their momentum and preserve their army to reach Moscow (and maybe beyond). Each mission represents a battle Napoleon's army had to endure in the heart of Russia, so the story follows the places where the biggest battles were fought.


Napoleon's Russian Campaign, 1812
422,000
June 1812
Napoleon and his troops march across the Neman River and into Russia.
50,000
Napoleon sends troops to Polotsk to protect his left flank.
175,000
Reduced by desertion, disease, starvation, and capture, an army of 175,000 arrives in Smolensk. Another 30,000 die there.
130,000
Sept. 7, 1812 Napoleon's army fights the Battle of Borodino and suffers 30,000 casualties.
Sept. 14, 1812 Napoleon enters Moscow to find it in ashes, torched by the czar. He waits, hoping to induce the czar to surrender.
Oct. 18, 1812 Frustrated and starving, having waited too long for the czar, the 100,000 survivors of the Grand Army begin their hellish retreat through the cruel Russia winter.
November 1812
The army returns to Smolensk and finds famine. The remaining 24,000 march on, abandoning their wounded.
37,000
Dec. 6, 1812
Troops march for the Neman River. Only 10,000 make it out of Russia.
28,000
The 30,000 in Polotsk join the 20,000 survivors. Thousands drown while crossing the Berezina River.
50,000
RUSSIA
Western Dvina River
Moscow R.
Oka R.
Moscow
Borodino
Maloyaroslavets
Vyazma
Polotsk
Vitebsk
Smolensk
Glubokoye
Kovno
Vilna
Molodechno
Borisov
Minsk
Dnieper River
Berezina River
Neman River
PRUSSIA
GRAND DUCHY OF WARSAW
N
0 100 Miles
0 200 Kilometers
Advancing troops
Retreating troops
= 10,000 soldiers
= 10,000 lost troops
GEOGRAPHY SKILLBUILDER: Interpreting Maps
1. Movement How long did it take the Grand Army to cover the distance between the Russian border and Moscow?
2. Place Why was it a mistake for Napoleon to stay in Moscow until mid-October?

Appendix A – Power, Morale and the army

Back to overview: 3 Power and Morale

A.1 Power

- Represents the strength of an army, including its troops and equipment.
- It is measured as a whole number.
- Power can be increased by interacting with points of interest distributed across the map.
- During battles, Power decreases automatically; it can be restored by resting after a battle.
- Enemy armies have their own Power, which can be reduced through points of interest on the map and/or through actions by specific agents.

A.2 Morale

- Represents the army's mood, fatigue and overall level of excitement.
- It is measured as a percentage in whole numbers (0%–100%).
- When Morale is below 50%, the army receives a penalty in battle; above 50%, it gains a bonus (see E.1 Battle formulas).
- Each turn, Morale decreases by 5% to represent the fatigue of marching. It does not decrease during battles or rest periods.
- Morale can also be increased by interacting with specific points of interest or through agents' actions.
- Enemy armies have their own Morale, which works like the player's, though they may receive higher bonuses from settlements. The player can reduce it with agents' actions at points of interest.

A.3 Army orders

Back to overview: 4 Army controls

Rest

After a battle, the player can let the army rest to recover Power and Morale. At the start of each round after a battle, a full-screen pop-up asks whether the army should march to the next objective or rest. Rest is limited to a fixed number of turns (TBD), and the remaining turns must be clearly visible in the UI. When the counter ends, if the player has not moved yet, a pop-up warns that the army will start moving at the end of the round.

Advance

After a battle, the player can start the army's march early. This stops the benefits of resting, but can be useful to reach the next objective before the enemy reinforces it. If the player chooses to march in the same full-screen pop-up, the army starts moving toward the next objective at the end of the turn.

Choose a destination

When the army reaches a crossroads (a cell with a crossroads sign), the player must choose the next objective between two clearly separated points of interest. Each objective grants a specific bonus. The army cannot go backward, so once a path is chosen the other objective can no longer be reached. At the start of that turn a pop-up asks where to go next, and the player must choose before moving on; at the end of the turn the army moves toward the chosen objective.

Appendix B – Agents

Back to overview: 5 Agents

B.1 Armed band

A band of soldiers dedicated to disrupting enemy plans behind enemy lines.

- It has a level: each action gives experience, the experience needed grows with each level, and the level has a cap. The higher the level, the more effective its actions (e.g. fewer turns, stronger enemies defeated).
- It can fight enemy armed bands: if it wins, the enemy band is destroyed; if it loses, it is destroyed itself (but this can slow down enemy actions).
- If it joins the main army, it increases the army's Power.
- It cannot move far: 2 cells per turn.

B.2 Spy

A unit dedicated to infiltrating behind enemy lines, gathering information and sabotaging enemy plans.

- It has no Power or Morale.
- It has a level: each action gives experience, the experience needed grows with each level, and the level has a cap. The higher the level, the more effective its actions (e.g. fewer turns, better rewards).
- It can infiltrate enemy armed bands, gather information from inside and mark them on the map while it stays with them.
- If it joins the army, it raises the army's Morale.
- It can move quite far: 3 cells per turn.

B.3 Scout

A unit dedicated to spotting enemies on the map (armed bands and the enemy army), showing the player what is happening in the fog of war.

- It has a large vision range that clears the fog of war for a set number of cells around it.
- It has no Power or Morale.
- It has a level: each action on something it is spotting gives experience, the experience needed grows with each level, and the level has a cap. The higher the level, the larger its vision range, and hidden elements become visible.
- It can perform a quick attack on armed bands: it does not destroy them, but lowers their Power and Morale.
- If it joins the army, it increases the army's vision range and, during battles, spots a crucial element each turn that, if destroyed, lowers the enemy's Morale or Power.
- It is fast: 4 cells per turn.

Appendix C – Points of interest

Back to overview: 6 Points of interest

C.1 Weapon deposit

An enemy armory where weapons, explosives, machines and gadgets are stored.

- **Armed band:** raids it in 1 turn: +100 Power for the army.
- **Spy:** sabotages the structure: the big settlement loses 100 Power.
- **Scout:** gathers weapons in 2 turns and delivers them to the army automatically: +120 Power for the army.

C.2 Rations deposit

An enemy food depot with the supplies used to replenish enemy forces.

- **Armed band:** raids it in 1 turn: +10% Morale for the army.
- **Spy:** poisons the food: the big settlement loses 10% Morale.
- **Scout:** gathers food in 2 turns and delivers it to the army automatically: +12% Morale for the army.

C.3 Bridge

A bridge over a river or a tunnel that lets agents and the army cross.

- **Armed band:** hides beneath the bridge to set up an ambush that:
  - destroys an enemy armed band, whatever its level, with no risk;
  - or reduces the enemy army's Power and Morale depending on the armed band's level; this destroys the armed band.
- **Spy:** sets traps on the bridge: the next enemy to cross loses 30 Power and 10% Morale and stops for 1 turn.
- **Scout:** rigs the bridge and destroys it, so nobody can cross. It cannot do this while another agent is on the bridge.

C.4 Small settlement

Small villages scattered across the map, never on the army's main path. They are small enough to be conquered by agents alone.

- Settlements have levels that decide how hard they are to conquer. An armed band can conquer a settlement only if its level is equal to or higher than the settlement's; if the settlement's level is higher, the armed band is destroyed.
- When an armed band is stationed in the settlement, its level is added to the settlement's, protecting it from future attacks.
- Under enemy control, the settlement can generate enemy armed bands.

Once conquered, the player chooses what it produces each turn

- **Create resources:** +5 Power and +2% Morale for the army every turn.
- **Recruit:** an armed band, spy or scout, depending on the settlement, up to a fixed number per settlement. Each settlement can recruit only one type of unit.
- **Reinforce:** takes 2 turns; the passive bonus stops meanwhile. The settlement goes up one level (up to level 4): each level adds +5 Power and +1% Morale to the passive bonus, up to +20 Power and +5% Morale each turn. It also becomes harder for the enemy to conquer.

Agents' actions and bonuses

- **Armed band (not conquered):** attacks the settlement directly. If the settlement's level is lower than or equal to the band's, it is conquered in 1 turn and stops giving bonuses to the

enemy army. Taken this way, it gives no passive bonus for 3 turns and its production starts again from level 1.

- **Armed band (conquered):** stationed in the settlement, it greatly increases the settlement's defense against enemy bands.
- **Spy (not conquered):** converts the settlement to the Napoleonic cause in 3 turns and keeps its level of passive bonus; the enemy's bonus stops only on the 3rd turn.
- **Spy (conquered):** stationed in the settlement, it adds +2% Morale to the passive bonus.
- **Scout (not conquered):** cannot conquer the settlement, but its guerrilla actions lower its level. At level 0 the settlement is conquered by an armed band without a fight.
- **Scout (conquered):** stationed in the settlement, it adds +5 Power to the passive bonus.

C.5 Medium settlement

Medium-sized villages that are strategic for the conquest of the region. They are always on the army's path and cannot be conquered without it.

- Settlements have Power and Morale values that decide how hard they are to conquer.
- An enemy army can station itself in the settlement and add its Power and Morale to it.
- Under enemy control, the settlement generates Power and Morale for the enemy; the bonus stops as soon as the attack starts.
- Once conquered, if it is not being raided, the army can station itself there for a set number of turns to rest and recover Power and Morale after a battle. While stationed, the army's Power and Morale are added to the settlement's.

Once conquered, the player chooses what it produces each turn

- **Create resources:** +20 Power or +4% Morale (base) for the army every turn.
- **Training:** agents rest and gain experience as a passive bonus, levelling up and becoming more efficient on the battlefield.
- **Raid:** just after the conquest, if the player wants to push forward, the army raids the settlement: +75 Power and +10% Morale at once, but no passive bonus for 4 turns. The army starts marching to the next objective at once.
- **Reinforce:** increases the settlement's production each turn: +10 Power and +2% Morale per level, up to level 4. The passive bonus stops while it happens.

Agents' actions

These actions are possible only when the army is not in the same cell; if it is, agents only boost the army.

- **Armed band (not conquered):** cannot interact with the settlement. If it tries to enter, it is destroyed in a kamikaze attack that reduces the enemy army's Power and Morale based on the band's size.
- **Spy (not conquered):** gathers information about nearby enemy agents and points of interest, and reveals the Power of enemy armies.
- **Any agent (conquered):** trains by staying in the settlement, to become more efficient on the battlefield.

C.6 Big settlement

Cities that represent the final step in conquering the region. They are always on the army's path and cannot be conquered without it. Conquering one ends the match.

- Settlements have Power and Morale values that decide how hard they are to conquer. In the prototype the big settlement starts with 700 Power and 85% Morale.
- An enemy army can station itself in the settlement and add its Power and Morale to it.
- Under enemy control, the settlement generates Power and Morale for the enemy.

Agents' actions

These actions are possible only when the army is not in the same cell; if it is, agents only boost the army.

- **Armed band:** cannot interact with the settlement. If it tries to enter, it is destroyed.
- **Spy:** gathers information about nearby enemy agents and points of interest.
- **Scout:** cannot interact with the settlement. If it tries to enter, it is destroyed.

C.7 Artillery position

An enemy position (level 2 by default) near a battle or settlement. It boosts the Power of nearby enemy armies or settlements; if it is destroyed or sabotaged, the bonus is removed.

- **Armed band:** attacks and destroys the position in 1 turn: the big settlement loses 50 Power.
- **Spy:** sabotages the position: the big settlement loses 25 Power each turn for 3 turns, then the position is destroyed.

C.8 Command camp

The enemy general's camp (level 2 by default), near a battle or settlement. It boosts the Morale of nearby enemy armies or settlements.

- **Armed band:** attacks the camp and kills the general: the big settlement loses 10% Morale.
- **Spy:** spreads disorder in the camp: the big settlement loses 5% Morale each turn for 3 turns.

Appendix D – Enemies

Back to overview: 7 Enemies

The following are examples of the actions and behaviors of enemy agents on the map.

D.1 Russian armed band

Tasked with eliminating the player's agents and with defending or reconquering points of interest.

- Ambushes on bridges or in forests, to eliminate the player's armed bands or slow the advance of Napoleon's army.
- Encamps in a settlement to reinforce its defenses and make it harder to conquer.
- Captures a Napoleonic settlement: its passive Power and Morale income stops and goes to the enemy army.
- Takes out an armed band if its own level is higher.
- Takes out a spy or scout if other agents have revealed it.
- Joins the Russian army to boost its Power.

D.2 Russian spy

Gathers information for the AI and lowers the player's Power and Morale income. The player cannot see it unless information about it is discovered (which makes it permanently visible on the map) or a scout is nearby.

- Cloaks itself: it stops its current action, but lower-level scouts cannot detect it.
- Sabotages a weapon or rations deposit, reducing what the player can gather from it each turn. If not stopped, it eventually destroys the deposit.
- Infiltrates a Napoleonic armed band to gather information about other agents nearby.
- Infiltrates a Napoleonic settlement, reducing its passive bonuses and slowing its upgrades.
- Infiltrates the Napoleonic army to reduce its Morale each turn.

D.3 Russian scout

Reports the player's movements to the AI and slows down Napoleonic agents and the army. The player cannot see it unless information about it is discovered (which makes it permanently visible on the map) or a scout is nearby.

- Discovers nearby scouts and spies of a lower level.
- Joins an artillery position, boosting the Power of the nearby Russian army or settlement.
- Rigs a weapon or rations deposit: the agent that interacts with it is not killed, but is stunned for one turn.
- Rigs bridges, stunning anyone who crosses them (Napoleon's army included).
- Infiltrates a Napoleonic settlement, improving its vision of the area and detecting Napoleonic agents near it more easily.
- Infiltrates the Napoleonic army to reduce its Power each turn.

D.4 Russian sniper

Defends specific areas and cells, and can attack only within a cone of vision. Until it fires, it is invisible to all units (unless a spy gathers information about it).

- Its first shot does not kill: the agent flees from the cell and loses all its moves. The next shot makes it flee again and lowers its level by one; an agent already at level 1 dies.
- Defends 3 cells: every enemy entering the area is shot on sight.
- Defends a road and shoots at the army, reducing its Power and Morale with each shot.
- Repositions to become invisible again (not if a spy has already gathered information about it).

D.5 Russian Dog Master

Captures and kills Napoleonic agents with two groups of dogs that roam freely around him.

- A group of dogs that finds an agent stuns it for the next turn. On the following turn, the Dog Master and the other group reach the agent and start a fight.
- Dogs can be killed if an agent reaches them during the player's turn. If the Dog Master is killed, the dogs become strays and attack every agent they meet, Russian ones included.
- Sends dogs to search specific places for agents.
- Trains new dogs if one or both groups are killed.
- Recalls his dogs if he finds the agent himself; this boosts his level in the fight.

D.6 Russian trapper

Blocks, ambushes and reports Napoleonic agents to other Russian agents. The player cannot see it unless information about it is discovered (which makes it permanently visible on the map) or a scout is nearby.

- Places traps that only scouts can see. A trap blocks an agent for one turn and reveals its position to all Russian agents.
- Ambushes agents from the forests: it does not kill them but lowers their level, then flees and becomes invisible again.
- Places traps on the main road that stun the army for one turn; this also stops the passive income from settlements.

D.7 Russian army

The enemy army. More than one can be on the map at the same time. It has Power and Morale and moves one cell per turn.

- Its task is to defend settlements and defeat the player's army. Each turn the AI looks at which target is nearer and which option gives the best chance of victory, and decides whether to attack the player's army or join a nearby settlement to fortify it.
- Only Napoleon's army can defeat it in battle. The battle takes several turns, depending on the difference in Power and Morale between the armies.
- Its position is always visible on the map, although it may spawn as reinforcements after the match has begun.

Agents' actions against it

- **Armed band:** attacks the army directly. It is a suicide mission, but the enemy loses 50 Power and 5% Morale and stalls for 1 turn.
- **Spy:** infiltrates the army: −2% Morale each turn.
- **Scout:** quick hit-and-run attacks: −20 Power each turn.

D.8 Prototype armed band behavior

Back to overview: 7.1 Prototype: Russian armed band

In the prototype, when an agent conquers a specific settlement, particularly a medium or big one, a Russian armed band spawns and tries to retake it:

- **1.** The player's agent conquers a specific point of interest. This settlement must have an action that triggers the spawn once, upon conquest.
- **2.** An armed band spawns from a specified medium or big settlement and moves toward the point of interest.
- **3.** If the armed band is met and defeated before it reaches the point of interest, nothing else happens.
- **4.** If the armed band reaches the point of interest, it reconquers it in 1 turn, whatever the settlement's level. No other action can happen in that cell during the conquest.

- **5.** Once the point of interest is reconquered, every bonus it gave the army is lost and the settlement goes back to its original level.
- **6.** The Russian armed band is destroyed and does not leave the point of interest, as if it became part of it.
- **7.** The player must conquer the point of interest again to get back the army bonus.

Appendix E – Battle formulas and prototype values

E.1 Battle formulas

Back to overview: 9 Battle system

The Health of an army is a percentage of its total Power. Every turn, each army deals a certain amount of Damage per turn to the other. The battle is won when the enemy army has no Health left or is routed.

Symbol	Meaning
P1, M1	Power and Morale of the player's army
P2, M2	Power and Morale of the enemy army

Damage factor (DF)

$$DF = abs(P1 - P2)$$

Power ratio (PR)

$$PR1 = P1 / (P1 + P2)$$

$$PR2 = P2 / (P1 + P2)$$

Damage per turn (DPT)

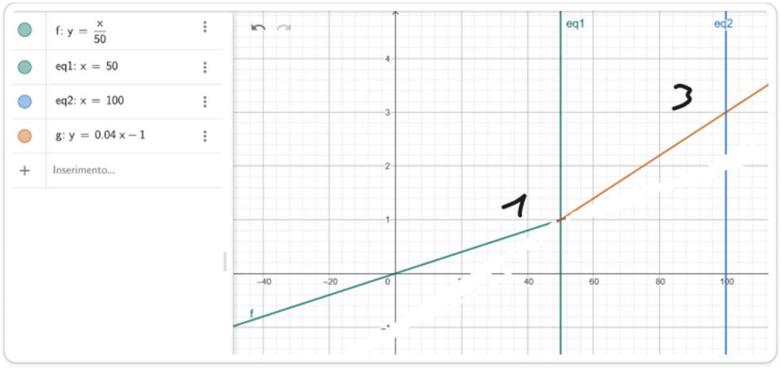
$$DPT1 = DF * PR1 * f(M1) = abs(P1 - P2) * P1 / (P1 + P2) * f(M1)$$

$$DPT2 = DF * PR2 * f(M2)$$

The damage inflicted on the other army each turn of the battle. It is NOT shown and does not reduce the current Power of either army.

f(M) uses Morale as a whole number (M × 100). It is 1 or less when Morale is 50% or lower (penalty) and more than 1 when Morale is above 50% (bonus), up to 3 at 100% Morale:

$$f(M1) = \begin{cases} M1/50 \text{ if } M1 < 50 \\ 0.04 * M1 - 1 \text{ if } M1 \geq 50 \end{cases}$$



Probability of rout

$$p1 = (e^{\wedge}(k1 * D1/P1) - 1) / (e^{\wedge}k1 - 1))$$

$$k1 = 1 + M1 * 6$$

Where D is the damage taken since the beginning of the battle.

E.2 Prototype values

Back to overview: 3 Power and Morale

Starting values

Value	Prototype
Napoleon's army: starting Power	300
Napoleon's army: starting Morale	70%
Napoleon's army: Morale lost each turn of march	5%
Big settlement: starting Power	700
Big settlement: starting Morale	85%
Artillery position and command camp: starting level	2

Rewards

Point of interest	Action	Effect
Weapon deposit	Conquest by armed band	+100 Power to the army
Weapon deposit	Conquest by scout	+120 Power to the army
Weapon deposit	Sabotage by spy	−100 Power to the big settlement
Rations deposit	Conquest by armed band	+10% Morale to the army
Rations deposit	Conquest by scout	+12% Morale to the army
Rations deposit	Sabotage by spy	−10% Morale to the big settlement
Small settlement	Passive income	+5 Power and +2% Morale to the army each turn
Small settlement	Each upgrade (up to level 4)	+5 Power and +1% Morale more each turn
Small settlement	Maximum passive income	+20 Power and +5% Morale each turn
Artillery position	Conquest by armed band	−50 Power to the big settlement
Artillery position	Sabotage by spy	−25 Power to the big settlement each turn, for 3 turns
Command camp	Conquest by armed band	−10% Morale to the big settlement
Command camp	Sabotage by spy	−5% Morale to the big settlement each turn, for 3 turns